

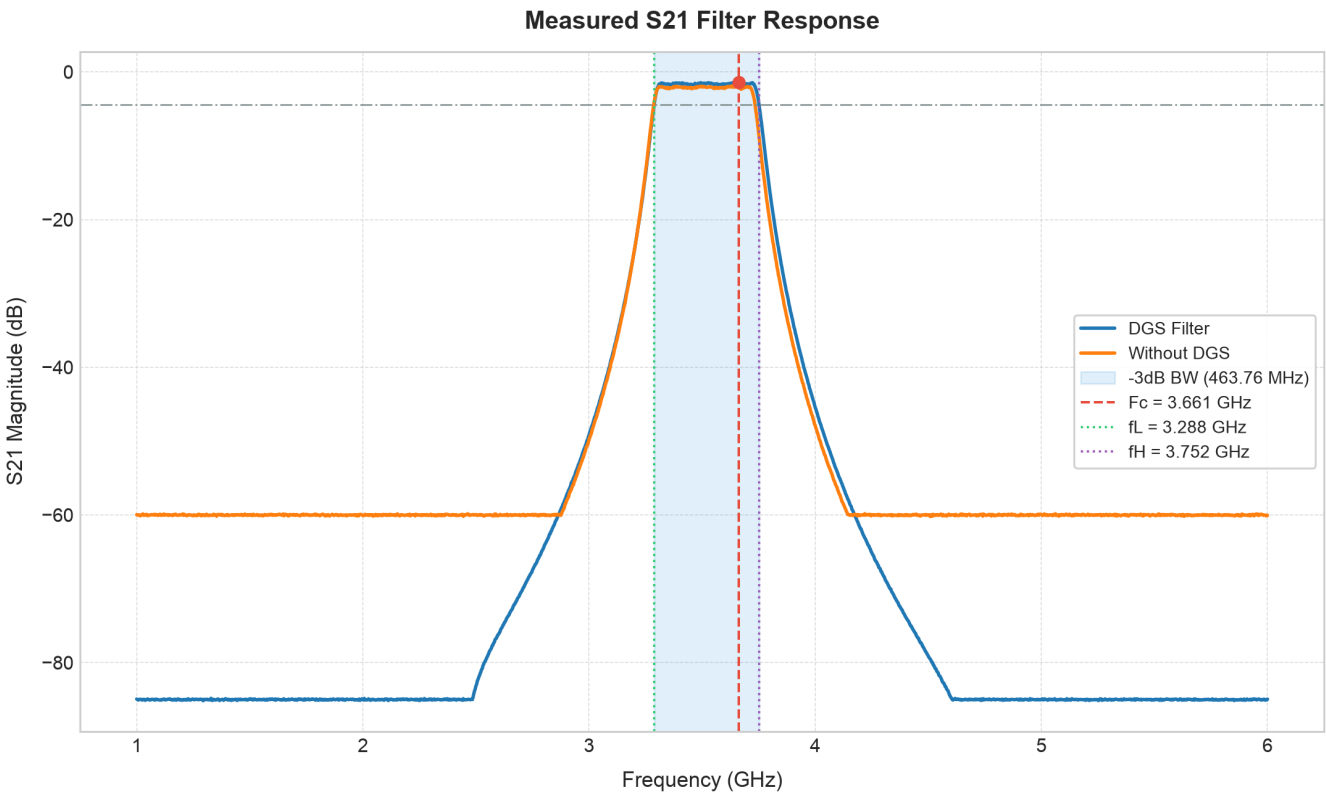
# Rohde & Schwarz VNA Filter Analysis Report

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## 1. Measurement Information

| Parameter           | Value                                         |
|---------------------|-----------------------------------------------|
| VNA Instrument      | Rohde & Schwarz ZNB20 Vector Network Analyzer |
| Active Sweep Points | 1601                                          |
| Frequency Range     | 1.000 GHz to 6.000 GHz                        |
| Measured Parameter  | S21                                           |
| Source File         | dgs_filter_5th_order.csv                      |
| IF Bandwidth        | 10 kHz                                        |
| Test Power Level    | -5 dBm                                        |

## 2. Interactive Frequency Response Plot



## 3. Primary Filter Characterization

| Parameter             | Calculated Value | Description / Formula                  |
|-----------------------|------------------|----------------------------------------|
| Center Frequency (Fc) | 3.661 GHz        | Frequency at maximum transmission peak |
| Peak S21              | -1.43 dB         | Maximum transmission magnitude         |
| Insertion Loss (IL)   | 1.43 dB          | -Peak S21 (magnitude of attenuation)   |

|                            |            |                                                       |
|----------------------------|------------|-------------------------------------------------------|
| Lower Cutoff (fL, -3dB)    | 3.288 GHz  | Interpolated lower threshold crossing                 |
| Upper Cutoff (fH, -3dB)    | 3.752 GHz  | Interpolated upper threshold crossing                 |
| Bandwidth (BW, -3dB)       | 463.76 MHz | fH - fL                                               |
| Fractional Bandwidth (FBW) | 12.67 %    | $(BW / F_c) \times 100\%$                             |
| Loaded Q Factor            | 7.90       | $F_c / BW$ (-3 dB bandwidth)                          |
| Passband Ripple            | 2.62 dB    | Peak-to-peak variation inside passband                |
| Min In-Band S11            | -27.46 dB  | Best return loss reflection point                     |
| Return Loss (RL)           | 27.46 dB   | -Min(S11) in positive dB convention                   |
| VSWR @ Fc                  | 1.23       | $(1 +  \Gamma ) / (1 -  \Gamma )$ at center frequency |

4. Multi-Measurement Comparison

| Parameter                  | DGS Filter | Without DGS |
|----------------------------|------------|-------------|
| Center Frequency (Fc)      | 3.661 GHz  | 3.631 GHz   |
| Peak S21                   | -1.43 dB   | -1.88 dB    |
| Insertion Loss             | 1.43 dB    | 1.88 dB     |
| Lower Cutoff (fL)          | 3.288 GHz  | 3.284 GHz   |
| Upper Cutoff (fH)          | 3.752 GHz  | 3.736 GHz   |
| Bandwidth (BW)             | 463.76 MHz | 452.54 MHz  |
| Fractional Bandwidth (FBW) | 12.67 %    | 12.46 %     |
| Loaded Q (Fc / BW)         | 7.90       | 8.02        |
| Passband Ripple            | 2.62 dB    | 2.84 dB     |
| Min In-Band S11            | -27.46 dB  | -22.30 dB   |
| Return Loss (RL)           | 27.46 dB   | 22.30 dB    |
| VSWR @ Fc                  | 1.23       | 1.27        |

DGS Comparative Summary:

- Measured center frequency shift: 30.504 MHz higher (3.631 GHz → 3.661 GHz).
- Measured insertion loss improvement: 0.45 dB (DGS Filter has 1.43 dB vs. Without DGS 1.88 dB).
- Measured bandwidth: 11.22 MHz wider (+2.5% change, 452.54 MHz → 463.76 MHz).
- Measured fractional bandwidth difference: +0.20% (12.46% → 12.67%).
- Measured return loss improved by 5.17 dB (22.3 dB → 27.5 dB).

5. Engineering Notes & Observations

Measured 5th-order hairpin filter test run.

**Engineering Notice & Disclaimer:** Reported values are derived from imported VNA measurement data via linear/spline interpolation. S-parameter parameters, Loaded Q, and cutoff boundaries are calculated metrics and should be verified with calibrated laboratory reference standards.